

GPU

MIG Instance A (12 SMs)

MIG Instance B (8 SMs)

$\updownarrow$ *in place*

Serving Processes

Publisher

Weights

KV Cache

model

prefix

private

Agent 1

Weights

KV Cache

model

prefix

private

Agent 2

Weights

KV Cache

model

prefix

private

...

①

②

③

Host Page Tables (SMMU)

$N : 1$

$N : 1$

$1 : 1$

Model File

Prefix Cache File

Private Extents

Immutable Extents

State File (Metadata)



Unified Memory

$\rightarrow$ Immutable, Shared

$\rightarrow$ Private, Writable

① Publish

② Attach

③ Grow Extent